A key feature of PVA is the definition of minimum viable population sizes and minimum habitat areas—information which can then inform decisions on conservation priorities.

A tool for conservationists

PVA combines both statistics and ecology to calculate the fewest organisms required for a species to survive long-term in its preferred habitat. This minimum number also dictates the amount of suitable habitat that the species needs. PVA is useful for conservationists when lobbying governments and developers to give protected status to an area. Armed with a PVA, they can explain precisely why reducing a stretch of forest, heathland, or reedbed will threaten certain flora or fauna. Protecting an area that is extensive enough to support a large species also benefits many smaller organisms sharing the same environment.

A number of creatures can only survive in environments where human disturbance is minimal. This is especially true for those that live in specialist habitats, such as certain owls in old-growth forest, reptiles on acid heathland, or amphibians in fast-flowing, unpolluted streams. However, as the human population grows, there is a constant demand for land for building, agriculture, leisure, roads, or forestry. This pressure is a particular threat to species that cannot easily adapt and move elsewhere. Where they are already confined to “islands” of suitable habitat, it takes no more than a low level of environmental damage or human disturbance to nudge them toward extinction.